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ROLL: Robust Noisy Pseudo-label Learning for Multi-View Clustering with Noisy Correspondence

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Abstract

Multi-view clustering (MVC) aims to exploit complementary information from diverse views to enhance clustering performance. Since pseudo-labels can provide additional semantic information, many MVC methods have been proposed to guide unsupervised multi-view learning through pseudo-labels. These methods implicitly assume that the predicted pseudo-labels are predicted correctly. However, due to the challenges in training a flawless unsupervised model, this assumption can be easily violated, thereby leading to the Noisy Pseudo-label Problem (NPP). Moreover, these existing approaches typically rely on the assumption of perfect cross-view alignment. In practice, it is frequently compromised due to noise or sensor differences, thereby resulting in the Noisy Correspondence Problem (NCP). Based on the above observations, we reveal and study unsupervised multi-view learning under NPP and NCP. To this end, we propose Robust Noisy Pseudo-label Learning (ROLL) to prevent the overfitting problem caused by both NPP and NCP. Specifically, we first adopt traditional contrastive learning to warm up the model, thereby generating the pseudo-labels in a self-supervised manner. Afterward, we propose noise-tolerance pseudo-label learning to deal with the noise in the predicted pseudo-labels, thereby embracing the robustness against NPP. To further mitigate the overfitting problem, we present robust multi-view contrastive learning to mitigate the negative impact of NCP. Extensive experiments on five multi-view datasets demonstrate the superior clustering performance of our ROLL compared to 11 state-of-the-art methods.

1. Introduction

In real-world applications, multi-view data [1] is typically collected from various sources, sensors, or feature extractors, which is widely used for a range of downstream tasks, such as classification [2, 3], clustering [4, 5], and retrieval [3, 6, 7]. As an important unsupervised technique, multi-view clustering (MVC) [8–10] aims to learn a common space to capture consistent and complementary information from all views, thereby grouping data into different clusters. In recent years, MVC has garnered extensive attention and has become a hot field of research. Numerous MVC methods [11] have been proposed, which can be broadly categorized into four types, i.e., subspace learning [12], non-negative matrix factorization [13], graph learning [14], and deep learning [15]. Although they have achieved promising progress, there are still some challenges in mitigating the semantic gap between multi-view heterogeneous data due to the lack of semantic supervision information [16, 17].

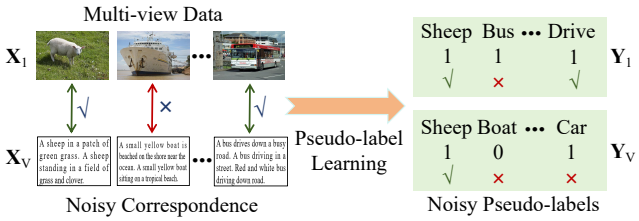


Figure 1. The motivation of our ROLL. Cross-view sample pairs are contaminated, which could cause some unaligned sample pairs to be mistaken for positive pairs, *i.e.* noisy correspondence problem. Existing pseudo-label guided MVC methods struggle to ensure flawless predictions, which inevitably leads to a lot of noise in the generated pseudo-labels, *i.e.* noisy pseudo-labels problem.

As an emerging technique, pseudo-label learning [18] has garnered significant attention in the field of MVC [19]. By automatically generating pseudo-labels for unlabeled multi-view data, pseudo-labels could provide the supervised information to guide the clustering process, thereby alleviating the semantic information scarcity issue inherent in traditional unsupervised learning approaches. The introduction of pseudo-labels not only facilitates capturing la-

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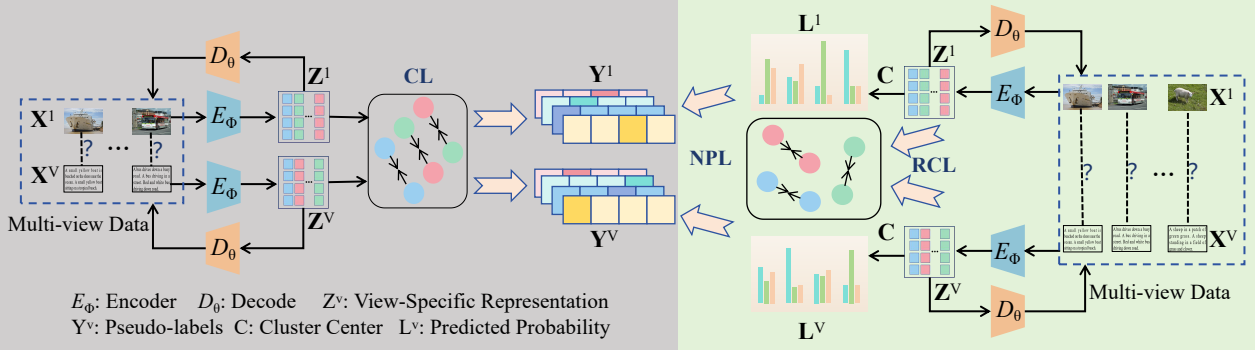


Figure 2. The framework of our ROLL. During the warm-up stage, we first obtain pseudo-labels of each view by contrastive learning (CL). Then, we conduct learning from noisy pseudo-labels through noise-tolerance pseudo-label learning (NPL) to robustly train the model under noisy pseudo-labels. Moreover, to further prevent the model from overfitting false correspondences, we propose robust multi-view contrastive learning (RCL) to effectively overcome the correspondence challenge.

tent semantic correlations between multi-view data to reveal their intrinsic structures, but also can provide valuable prior knowledge, such as cluster centers or boundaries. For example, PGCL [20] proposes a new pseudo-label guided contrastive learning scheme, which iteratively leverages pseudo-labels to guide the removal of false negative sample pairs, thereby endowing the discriminability of contrastive learning.

However, almost all pseudo-label guided MVC methods implicitly assume that the predicted pseudo labels are correct. In practice, due to the difficulty of learning perfect labels from unlabeled data, they cannot guarantee accurate estimates, which inevitably leads to the noisy label problem, as shown in Fig.1. In other words, the predicted pseudo-labels could have some errors and introduce a lot of noise, i.e., noisy pseudo-labels problem (NPP). Some recent works have revealed that neural networks easily overfit noisy labels, thereby resulting in inevitable performance degradation. In addition to the NPP, a further limitation in existing MVC methods is their strong reliance on the assumption of perfect cross-view alignment. However, such an assumption is difficult or even impossible to satisfy. In practical scenarios, this inevitably leads to the contamination of the cross-view correspondence, as shown in Fig.1. In other words, some unaligned multi-view data may be incorrectly treated as aligned ones, i.e., noisy correspondence problem (NCP) [21]. Due to the false guidance, NCP will mislead the model, thereby remarkably degrading the clustering performance. Overall, when faced with a practical and more challenging scenario, the key issue of MVC lies in simultaneously mitigating the negative effects of noisy pseudo-labels and noisy correspondence problems.

To overcome the aforementioned problems, we propose a novel multi-view clustering paradigm with noisy correspondence, namely Robust nOisy pseudo-Label Learning (ROLL), which could mitigate the negative misdirection of both NPP and NCP. As shown in Fig.2, our proposed ROLL contains two stages, *i.e.* Pseudo-Label Learning (PLL) and Learning from Noisy Pseudo-Labels (LNPL). Specifically, we first warm up the model to obtain specific pseudo-labels for each view in a self-supervised learning manner, thereby endowing semantic discrimination. However, the generated pseudo-labels inevitably contain some noise. In addition, due to the existence of NCP, multi-view pseudo-labels could have inconsistent biases. To alleviate the overfitting problems [21], we design a novel noise-tolerance pseudo-label learning (NPL) loss that robustly learns from both NPP and NCP. Afterward, to achieve multi-view consistency, we propose robust multi-view contrastive learning to further alleviate the negative influence of false correspondences. In general, our major contributions are summarized as follows:

- We propose a novel robust noisy pseudo-label learning framework (ROLL) that robustly learns from both NPP and NCP. To the best of our knowledge, this work could be the first to reveal and study the noise problem in the pseudo-labels for unsupervised MVC.
- To alleviate the influence of the hard samples, we propose a novel noise-tolerance discriminative loss to deal with noisy pseudo labels, thus embracing robustness. We further design a robust multi-view contrastive loss to reduce the optimization attention on noisy pairs, thereby alleviating the model overfitting NCP.
- Extensive experiments show the effectiveness and robustness of our ROLL compared with 11 state-of-the-art MVC methods.

2. Related Work

2.1. Multi-view clustering

Over the past years, numerous MVC methods [22–24] have been proposed. For instance, SDMVC [25] adopts a self-supervised manner to construct a unified target distribution,

thereby overcoming the negative impact caused by certain views with unclear clustering structures. To provide more semantic information to guide unsupervised MVC [26], more and more pseudo-label learning methods have been proposed to utilize pseudo-labels to capture the semantic relevance between instances. For example, DuaLGR [18] utilizes global high-level class information from pseudo-labels to adaptively adjust the graph aggregation and to evaluate the view weight, thereby generating a consensus graph for MVC. To further improve the quality of representation learning, PE-MV-DCFCM [27] adopts pseudo-label learning to learn common representations between different views to preserve intra-view and inter-view similarities simultaneously. In addition, to deal with incomplete MVC tasks, PLP-IMVC [28] proposes a pseudo-label paradigm that integrates the pseudo-labels of observable data with deep label propagation networks to enhance clustering performance. Almost all pseudo-label guided MVC methods assume that the obtained pseudo-labels are faultless. However, it is impractical to guarantee perfect pseudo-label prediction, which inevitably introduces some noise. Such false supervision information could mislead the model, thus weakening the clustering performance. To this end, we propose a robust multi-view learning paradigm to alleviate the adverse impact of noisy pseudo-labels.

2.2. Contrastive Learning

As an important unsupervised learning paradigm, contrastive learning [29, 30] greatly improves the performance of representation learning. Fundamentally, this paradigm first builds a latent space and then maximizes the similarity between positive pairs while minimizing that between negative pairs. Positive and negative pairs are constructed from the invariant representations of all multi-view instances of the same sample and multiple views of different samples, respectively. With the popularity of contrastive learning, several contrastive learning methods for MVC have emerged. For example, to extract more refined feature representations, MFLVC [15] performs contrastive learning on high-level features and semantic labels to enhance the representation ability. To explore view-invariant representations, CVCL [31] proposes a cluster-level cross-view contrastive learning to capture consistent semantic label information between multi-view data. To maintain the consistency of samples that are similar yet distinct in cross-view scenarios, Dealmvc [32] proposes the dual contrastive calibration network to improve the discrimination of feature representation. To deal with a more practical noisy correspondence problem in MVC, RMCNC [21] proposes a noise-tolerance multi-view contrastive loss to reduce the influence of false negatives. The differences between the existing methods and our work are as follows. First, unlike most MVC methods that utilize pseudo-labels to provide semantic relevance between instances, we reveal and study the noise problem in the pseudo-labels for unsupervised multi-view data. Second, to the best of our knowledge, this work could be the first to learn with noisy pseudo-labels for robust MVC under noisy correspondence.

3. Method

3.1. Notations and Overview

Given a multi-view dataset $\left\{\boldsymbol{X}^v \in \mathbb{R}^{n\times d_m}\right\}_{v=1}^{V}$ with n instances across V views, $\boldsymbol{X}_i^v$ represents the i-th sample from the v-th view of multi-view data. Suppose that the number of clusters is K. The goal of MVC is to group unlabeled multi-view data with the same semantics into the same cluster. The pipeline of our proposed ROLL is illustrated in Fig.2, which mainly includes two key stages: Pseudo-Label Learning (PLL) and Learning from Noisy Pseudo-Labels (LNPL). Specifically, in the first stage, we warm up the model to obtain pseudo-labels, thereby providing semantic guidance for the next stage. In the second stage, we fine-tune the model by adopting the supervised information of pseudo-labels. However, learning semantic labels from unlabeled multi-view data is challenging. This inevitably leads to the problem of noisy pseudo-labels, thereby making the model overfit false labels. To this end, we propose Noise-tolerance Pseudo-label Learning (NPL) to estimate cross-view correspondence caused by NPP. In addition, to further alleviate the overfitting problem [21] caused by NCP, we propose robust multi-view contrastive learning to reduce the focus on noisy pairs.

3.2. Autoencoder Reconstruction

Deep autoencoders have been widely embraced in MVC due to their capability to learn clustering-friendly features. The encoder network aims at transforming multi-view data $\boldsymbol{X}^v$ into the view-specific representations $\boldsymbol{Z}^v \in \mathbb{R}^{n\times d}$ to explore underlying structure, *i.e.*

$$\boldsymbol{Z}^v = \boldsymbol{E}_{\phi^v}(\boldsymbol{X}^v), \tag{1}$$

where $\boldsymbol{E}_{\phi^v}$ and ϕ^v represent the encoder network modules and encoder parameters of the v-th view, respectively. To preserve as much as possible view-specific information and prevent the model collapse, we further adopt the following formula to optimal the reconstruction loss $\mathcal{L}_{rec}$ between input and output of all views, *i.e.*

$$\mathcal{L}_{rec} = \sum_{v=1}^{V} \left\|\boldsymbol{X}^v - \boldsymbol{D}_{\theta^v}\left(\boldsymbol{E}_{\phi^v}(\boldsymbol{X}^v)\right)\right\|_2^2, \tag{2}$$

where $\boldsymbol{D}_{\theta^v}$ and θ^v symbolize the decoder network modules and decode parameters of the v-th view, respectively.

3.3. Noise-tolerance Pseudo-label Learning

After the warm-up stage, we first learn a shared cluster center $\boldsymbol{C} \in \mathbb{R}^{c\times d}$ and the pseudo-labels $\boldsymbol{Y}^v$ for each view. Then, we could utilize the guidance information of the pseudo-labels to enhance the discriminative power of the learned representations. More specifically, we first calculate the predicted probability distribution $\boldsymbol{L}_i^v$ for the i-th sample in v-th view as follows:

$$\boldsymbol{L}_i^v = \frac{e^{\boldsymbol{Z}_i^v(\boldsymbol{C})^\top}}{\sum_j^K e^{\boldsymbol{Z}_j^v(\boldsymbol{C})^\top}}. \tag{3}$$

Then, we employ the traditional cross-entropy loss $\mathcal{L}_{ce}$ as the classification objective, which could be expressed as

$$\mathcal{L}_{ce}^v = -\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{Y}_i^v \log \boldsymbol{L}_i^v. \tag{4}$$

However, due to the inconsistency of cross-view correspondence and the absence of well-labeled ground truths, the classification results could be unreliable. In other words, the generated pseudo-labels may contain a lot of errors or noise. To this end, we propose noise-tolerance pseudo-label learning to alleviate the adverse impact of noisy pseudo-labels. Specifically, we design a cross-view correspondence weight to eliminate the correspondence relationship between different views. The loss function could be formulated as follows:

$$\mathcal{L}_{npl}^v = -\frac{1}{n}\sum_{u\neq v}^{V}\sum_{i=1}^{n} \frac{(\boldsymbol{L}_i^v)^\top \boldsymbol{L}_i^u}{\sum_{j=1}^K L_{ij}^v L_{ij}^u} \boldsymbol{Y}_i^v \log \boldsymbol{L}_i^v. \tag{5}$$

Further, the total noise-tolerance loss for all views can be written as follows:

$$\mathcal{L}_{npl} = \sum_{v=1}^{V} \mathcal{L}_{npl}^v. \tag{6}$$

3.4. Robust Multi-view Contrastive Learning

To reduce the cross-view heterogeneity gap in multi-view data, we adopt vanilla contrastive learning to maximize the consistency between positive pairs while minimizing the consistency between negative pairs. Specifically, we first use the cosine distance to measure the similarities between cross-view representations, *i.e*.

$$S(\boldsymbol{z}_i^v, \boldsymbol{z}_j^u) = \frac{\boldsymbol{z}_i^v(\boldsymbol{z}_j^u)^\top}{\|\boldsymbol{z}_i^v\|\|\boldsymbol{z}_j^u\|}, \tag{7}$$

where $\boldsymbol{z}_i^v$ and $\boldsymbol{z}_j^u$ are the view-specific representations. To capture cross-view consistency, we calculate the probabilities of the positive pair and the negative pairs, respectively. The probabilities can be written as

$$\begin{cases} (P_i^+)^{vu} = \exp(S(\boldsymbol{z}_i^v, \boldsymbol{z}_i^u)/\tau), \\ (P_i^-)^{vu} = \sum_{j\neq i, j=1}^{n} \exp(S(\boldsymbol{z}_i^v, \boldsymbol{z}_j^u)/\tau). \end{cases} \tag{8}$$

To learn discriminative representations, we adopt one of the most widely used contrastive loss InfoNCE, *i.e*.

$$\mathcal{L}_{cl}^{vu} = -\frac{1}{n}\sum_{i=1}^{n} \log \frac{(P_i^+)^{vu}}{(P_i^-)^{vu} + (P_i^+)^{vu}}, \tag{9}$$

where τ denotes a temperature parameter. The overall loss function across all views can be formulated as

$$\mathcal{L}_{cl} = \sum_{v=1}^{V}\sum_{u\neq v}^{V} \mathcal{L}_{cl}^{vu}. \tag{10}$$

However, as pointed out by [21], due to the memorization effect of DNN, InfoNCE pays too much attention to hard pairs, thereby leading to the overfitting problem. To alleviate the noise interference introduced by multi-view data with NC, we propose the following robust multi-view contrastive loss:

$$\mathcal{L}_i^{vu} = \frac{(1-r)\left((P_i^+)^{vu} + (P_i^-)^{vu}\right)^r - \left((P_i^+)^{vu}\right)^r}{r}, \tag{11}$$

where $0 < r < 1$ is a control factor. According to the robustness analysis, we could adjust the r value of the loss to enhance resistance against NC. The overall loss function can be represented as:

$$\mathcal{L}_{rcl} = \sum_{v=1}^{V}\sum_{u\neq v}^{V}\left(\frac{1}{n}\sum_{i=1}^{n}\mathcal{L}_i^{vu}\right). \tag{12}$$

3.5. Optimization

Our method contains two steps, *i.e*. Pseudo-Label Learning (PLL) and Learning from Noisy Pseudo-Labels (LNPL). Specifically, **PLL** aims to mine the preliminary semantic labels from unlabeled multi-view data by warming up the model. Specifically, we first adopt an autoencoder reconstruction loss $\mathcal{L}_{rec}$ to preserve view information. To learn the cross-view consistency, we utilize a vanilla contrastive learning loss $\mathcal{L}_{cl}$ (see Eq.10) to narrow the cross-view heterogeneity gap. The overall PLL loss $\mathcal{L}_{pll}$ could be expressed as follows

$$\mathcal{L}_{pll} = \mathcal{L}_{rec} + \lambda\mathcal{L}_{cl}, \tag{13}$$

where λ is a parameter that is empirically set to 1e-4. Further, we concatenate view-specific representations $\boldsymbol{Z}^v$ to obtain the representation $\boldsymbol{H} \in \mathbb{R}^{2n\times d}$ and adopt k-means clustering to obtain the shared cluster center $\boldsymbol{C} \in \mathbb{R}^{c\times d}$. To prevent the label inconsistency problem of multi-view data caused by NCP, we regard $\boldsymbol{C}$ as the initial cluster center to perform K-means clustering on each view, thereby obtaining pseudo-labels $\boldsymbol{Y} \in \mathbb{R}^{n\times c}$ of each view. **LNPL** aims to learn more discriminative representations under the guidance of the pseudo labels. To mitigate the negative impact

of noise from the pseudo-labels, we propose a novel noise-tolerance pseudo-label learning loss to reduce the focus on the hard samples. Moreover, to prevent the overfitting problem caused by NCP, we further propose robust multi-view contrastive learning to alleviate the negative influence of false correspondences between sample pairs. Thus, the overall loss $\mathcal{L}_{lnpl}$ could be formulated as follows

$$\mathcal{L}_{lnpl} = \mathcal{L}_{rec} + \alpha\mathcal{L}_{npl} + \beta\mathcal{L}_{rcl}, \tag{14}$$

where α and β are two trade-off hyper-parameters. Finally, we concatenate the corresponding view-specific representations to a common representation and perform K-means clustering to obtain the clustering results.

3.6. Discussion

For the proposed robust multi-view contrastive loss, when $r \to 1$, our loss could be rewritten as follows:

$$\mathcal{L}_i^{vu} = -(P_i^+)^{vu}. \tag{15}$$

Clearly, it is equivalent to the MAE loss [33]. MAE treats each sample equally, thereby leading to theoretical robustness against NC, which has been proven in RMCNC [21]. Although this equal treatment brings robustness, it inevitably loses focus on information pairs, thus resulting in the underfitting issue and performance degradation. When $r \to 0$, our loss could be expressed as follows:

$$\begin{aligned}
&\lim_{r\to 0} \frac{(1-r)\left((P_i^+)^{vu} + (P_i^-)^{vu}\right)^r - \left((P_i^+)^{vu}\right)^r}{r} \\
&= \lim_{r\to 0} \left(\frac{1 - \left((P_i^+)^{vu}\right)^r}{r} + \frac{-1 + \left((P_i^+)^{vu} + (P_i^-)^{vu}\right)^r}{r} \right) \\
&= \lim_{r\to 0} \frac{1 - \left((P_i^+)^{vu}\right)^r}{r} + \lim_{r\to 0} \frac{-1 + \left((P_i^+)^{vu} + (P_i^-)^{vu}\right)^r}{r} \\
&= -\log\left((P_i^+)^{vu}\right) + \log\left((P_i^+)^{vu} + (P_i^-)^{vu}\right) \\
&= -\log \frac{\left((P_i^+)^{vu}\right)}{\left((P_i^+)^{vu} + (P_i^-)^{vu}\right)} = \mathcal{L}_{\text{InfoNCE}}.
\end{aligned} \tag{16}$$

In general, in the limit of $r \to 0$, our loss becomes asymptotically equivalent to InfoNCE [29]. Intuitively, our loss mainly uses the control factor r to seek a balance between InfoNCE and MAE, which could focus less on hard samples than InfoNCE and more on informative samples than MAE, thus mitigating the negative impact of NC.

4. Expriments

4.1. Datasets

We perform expensive experiments to evaluate the proposed ROLL on five publicly multi-view datasets. Concretely, **Scene15** [34] comprises 4,485 images belonging to 15 classes, encompassing both indoor and outdoor environments. For each image, we extract 20-dim and 59-dim features by adopting GIST and PHOG, respectively. **CUB** [35] encompasses various birds, with the first ten categories selected for analysis. We employ deep visual features extracted from GoogLeNet and textual features obtained from doc2vec, representing two distinct views for our experiments. **Reuters** [36] contains news content in English and French with 18,758 samples. We use a conventional autoencoder to obtain 10-dimensional latent features of the texts. **NUS-WIDE** [37] consists of 9,000 instances with image-text pairs of 10 classes. Following [38], we adopt the 19-layer VGGNet to obtain the 4,096-dimension deep image features from the fc7 layer. We also adopt the sentence CNN to extract 300-dimensional text features. **XMediaNet** [39] comprises 40,000 instances with text and image pairs belonging to 200 classes. According to [38], we can get 4,096-dimensional deep image features and 300-dimensional text features.

4.2. Baselines and Experimental Setups

To verify the effectiveness of our method, we compare the proposed ROLL with 12 state-of-the-art MVC methods. They are MvCLN [40], MFLVC [15], SURE [41], DCP [42], DealMVC [32], CVCL [31], MSESC [43], RMCNC [21], ICMVC [44], SCM-RE [45], and MVCAN [46]. To comprehensively evaluate the performance on multi-view data with noisy correspondence, we regard the first view as the anchor view and randomly shuffle $m\%$ of the samples in other views, i.e., $m\%$ noise rate. In our setups, we vary noise ratios from 0%, 20%, 50%, to 80%. Then, we follow RMCNC [21] to perform the semantic-level alignment for the learned view-specific representations, thereby obtaining the realigned multi-view representations. Then, we concatenate these representations and perform K-means to achieve downstream MVC tasks. For the sake of fairness, we run all methods five times and record three average metrics (such as ACC, NMI, and ARI) of all comparison methods to evaluate the clustering performance. In general, the larger their values, the better the performance.

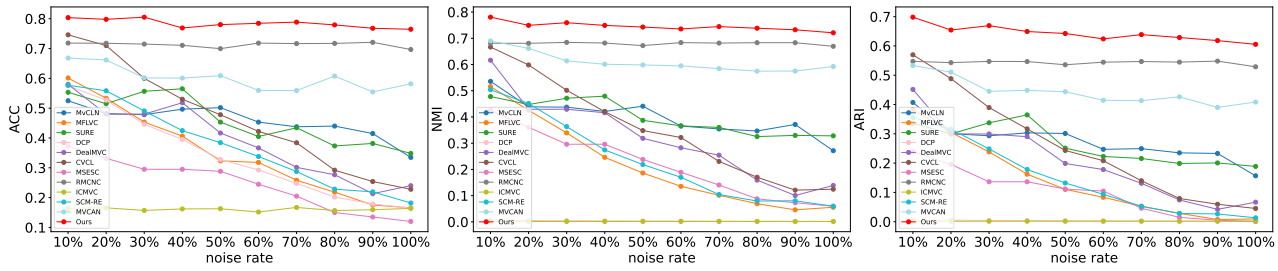
4.3. Implementation Details

The proposed ROLL[1] contains two stages. In the warm-up stage, the proposed network architecture contains a view-specific encoder-decoder, a linear layer, and a softmax layer. The fully connected architecture is adopted as the encoder-decoder for each view. Specifically, each encoder f_i has the same network structure, *i.e.* d_i-1024-1024-512, where d_i is the dimension of i-th view. Each decoder has the structure of 512-1024-1024-d_i. We use the Rectified Linear Unit (ReLU) as the activation function. The dimension of the linear layer is K, which is the number of classes. In the fine-tuning stage, we adopt the same encoder-decoder network architecture. Meanwhile, we standardize the max-

[1]The code is available at https://github.com/sunyuan-cs/2025-CVPR-ROLL.

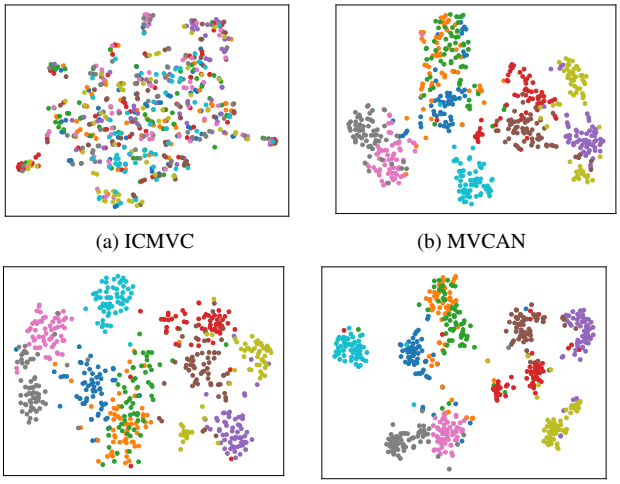
Table 1. The multi-view clustering performance (%) on five widely-used datasets with different noise rates. The best results are in bold. '/' represents out of memory.

Ratio	Methods	Scene-15			CUB			Reuters			NUS-WIDE			XMediaNet		
		ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
0%	MvCLN	39.60	41.60	25.10	55.67	53.58	38.97	45.30	22.75	21.02	58.52	43.38	37.26	21.82	67.07	15.23
	MFLVC	38.50	41.40	22.84	61.33	64.06	49.69	39.91	36.87	23.22	55.52	44.96	37.38	14.30	62.63	13.77
	SURE	40.95	43.19	25.01	63.17	57.47	43.67	48.09	28.57	25.38	56.30	46.01	38.60	51.39	82.92	25.29
	DCP	38.46	42.86	22.35	63.30	69.22	51.11	36.20	18.90	4.80	49.23	43.15	27.13	25.09	59.82	19.19
	DealMVC	35.61	40.19	22.06	63.00	71.86	54.85	36.85	32.99	19.18	64.29	50.45	45.52	7.20	53.79	7.97
	CVCL	42.97	42.62	24.97	81.20	76.36	68.39	53.15	32.08	25.28	61.14	47.12	41.50	21.22	57.27	16.26
	MSESC	36.47	40.07	21.59	42.50	51.25	31.18	41.51	29.77	10.75	39.71	38.62	25.08	/	/	/
	RMCNC	43.70	44.31	26.34	72.38	68.81	55.35	47.45	29.48	17.85	67.59	53.64	48.99	79.42	88.57	71.50
	ICMVC	38.77	41.11	23.06	30.93	33.91	16.47	50.07	32.58	25.12	18.72	8.07	3.63	/	/	/
	SCM-RE	40.83	38.89	22.62	68.93	66.51	52.77	37.68	27.91	17.88	59.12	49.65	39.20	80.33	**89.44**	74.31
	MVCAN	38.32	39.48	21.05	68.00	70.62	55.64	31.99	12.80	10.59	63.03	49.24	40.34	79.62	89.36	74.20
	Our ROLL	**47.14**	**48.76**	**30.54**	**82.90**	**83.16**	**74.79**	**50.76**	**37.91**	**28.34**	**71.06**	**57.62**	**53.33**	**80.79**	88.39	**74.55**
20%	MvCLN	36.45	35.92	21.11	48.17	43.86	30.04	29.97	12.55	6.57	52.36	36.79	31.27	21.09	59.74	16.24
	MFLVC	37.12	32.70	19.02	53.33	42.55	30.49	37.29	28.45	17.76	49.97	36.54	29.86	13.71	62.34	13.59
	SURE	35.67	32.27	17.91	51.50	44.75	29.97	41.40	17.95	12.58	48.69	35.84	29.19	42.18	72.51	27.23
	DCP	34.12	32.68	15.78	52.70	46.60	32.41	32.56	12.98	3.15	35.80	25.97	12.95	6.16	25.80	2.76
	DealMVC	27.94	26.48	14.98	56.67	55.20	38.93	37.98	30.38	19.33	56.54	36.29	32.83	5.58	40.75	5.44
	CVCL	34.16	33.63	18.51	71.00	59.84	48.82	47.91	26.06	20.10	52.59	38.03	31.94	19.46	47.96	12.66
	MSESC	32.17	28.14	15.05	33.17	36.06	19.56	30.83	19.01	4.92	36.14	34.35	19.61	/	/	/
	RMCNC	39.40	41.88	24.93	71.77	68.09	54.31	51.71	33.97	25.34	65.31	49.25	45.41	30.12	49.13	17.46
	ICMVC	37.84	33.58	19.93	16.57	3.91	0.44	46.74	25.24	20.48	12.29	0.94	0.25	/	/	/
	SCM-RE	32.74	25.06	14.75	53.90	44.45	31.13	33.48	20.46	13.30	57.67	47.29	37.09	80.79	88.89	74.13
	MVCAN	38.07	38.69	21.22	66.17	66.16	51.06	33.42	8.35	8.48	59.52	46.06	38.35	80.25	88.45	73.39
	Our ROLL	**45.65**	**45.00**	**28.15**	**81.03**	**76.81**	**68.11**	**51.94**	**36.51**	**29.08**	**66.71**	**50.61**	**46.60**	**81.28**	**89.19**	**75.62**
50%	MvCLN	34.08	33.38	17.41	50.17	44.10	30.12	39.02	18.33	14.03	34.87	17.27	12.56	13.53	40.77	6.90
	MFLVC	35.59	26.65	16.97	32.33	18.75	11.30	36.03	21.17	14.04	41.61	25.12	20.59	8.82	30.78	4.40
	SURE	38.82	32.26	20.30	45.33	38.72	25.12	32.47	13.67	10.27	36.32	18.60	13.49	39.41	63.69	27.40
	DCP	29.60	23.19	10.91	32.80	21.87	11.41	30.03	7.89	0.7	29.17	14.14	6.56	14.20	39.88	6.00
	DealMVC	22.74	17.93	8.32	41.67	31.86	19.88	43.03	13.69	11.83	39.11	16.06	13.10	4.91	22.27	2.92
	CVCL	32.04	25.56	14.68	47.80	34.81	24.33	44.83	18.24	15.92	44.29	29.68	24.29	3.51	15.42	0.93
	MSESC	23.05	16.48	7.60	24.50	18.90	7.95	23.79	10.92	0.51	39.41	29.38	21.14	/	/	/
	RMCNC	37.65	39.29	24.01	69.97	67.21	53.57	49.68	26.90	22.84	57.63	37.87	34.34	45.44	62.49	24.53
	ICMVC	29.94	22.14	12.27	16.27	2.78	0.09	41.87	15.79	13.44	12.14	0.54	0.17	/	/	/
	SCM-RE	21.49	11.09	5.36	37.10	21.00	12.49	29.01	12.67	9.17	56.02	41.50	31.53	80.93	87.85	73.69
	MVCAN	32.73	35.76	18.72	60.90	59.84	44.37	32.03	6.32	6.51	50.94	39.89	30.42	79.48	88.63	72.70
	Our ROLL	**45.00**	**43.05**	**27.24**	**77.63**	**73.60**	**63.80**	**52.44**	**33.13**	**27.59**	**58.97**	**42.45**	**35.70**	**82.24**	**89.43**	**76.52**
80%	MvCLN	23.28	24.01	10.97	44.00	34.66	23.52	32.59	16.95	14.15	23.76	10.61	5.93	9.64	35.74	4.70
	MFLVC	31.75	24.60	13.51	21.83	6.92	2.86	26.53	11.01	7.51	21.88	11.11	5.69	2.71	10.19	0.97
	SURE	20.07	24.10	9.28	37.33	32.52	19.88	38.30	17.89	15.84	27.90	11.87	8.09	21.26	48.57	12.00
	DCP	22.75	18.65	3.53	20.20	8.02	2.06	29.51	5.19	4.00	18.85	9.21	4.67	6.49	25.98	3.05
	DealMVC	23.72	22.84	11.22	27.67	15.92	7.42	32.12	8.65	4.32	26.31	5.36	3.50	2.45	6.15	0.29
	CVCL	32.13	22.16	13.29	29.20	17.02	7.93	36.92	11.14	10.07	35.73	20.69	15.10	6.67	19.41	1.13
	MSESC	15.65	9.28	3.35	15.00	8.79	1.49	24.01	6.10	1.73	13.12	4.18	1.56	/	/	/
	RMCNC	33.83	33.30	19.10	67.73	66.32	52.48	44.03	22.45	13.08	52.90	34.33	30.07	20.73	41.14	7.81
	ICMVC	23.47	11.50	5.90	15.66	2.92	0.09	31.26	6.71	5.83	12.02	0.58	0.15	/	/	/
	SCM-RE	15.33	5.38	2.40	23.80	8.82	3.41	25.87	12.34	8.31	41.41	27.49	16.81	79.64	86.12	71.23
	MVCAN	36.34	**37.52**	20.94	60.77	57.58	42.62	29.27	8.09	8.35	51.53	39.14	30.64	79.81	87.88	73.69
	Our ROLL	**39.84**	37.12	**21.74**	**78.40**	**74.14**	**63.80**	**46.25**	**26.85**	**18.89**	**54.81**	**39.37**	**31.00**	**81.87**	**88.73**	**75.54**



(a) ACC (b) NMI (c) ARI

Figure 3. The clustering performance of the CUB dataset under different noise rates.



(a) ICMVC (b) MVCAN

(c) RMCNC (d) Ours

Figure 4. The t-SNE visualizations on CUB with 50% noise rate

imum number of epochs $N_m = 150$ and warmup epochs $N_w = 50$ on all datasets, respectively. The batch size N_w and the temperature parameter τ are set to 128 and 0.5, respectively. For each dataset, the parameters (α and β) and the factor r are fixed as 1e-2, 1e-4, and 1e-1, respectively. To optimize the proposed ROLL, we adopt Adam Optimizer with a learning rate of 0.0001. All experiments are conducted using PyTorch 1.5.0 on an NVIDIA GTX 3090 GPU.

4.4. Comparisons with State-of-the-Arts

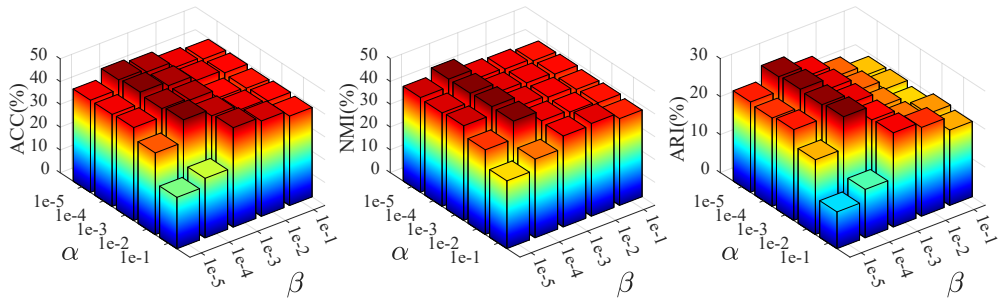
We perform extensive experiments to comprehensively show the clustering performance of our ROLL on six multi-view datasets with different noise rates. The experimental results are recorded in Tab.1. In addition, we draw the performance curves of all comparisons on CUB with different correspondence noise rates from 10% to 100% as shown in Fig.3. From these results, we can obtain the following observations: (1) In the complete scenario (i.e., 0% noise rate), the proposed ROLL achieves the best clustering performance on most datasets, compared with the state-of-the-art comparison methods. This is mainly attributed to the fact that the learned pseudo-labels provide more powerful semantic guidance. (2) When faced with NCP, the performance of most comparisons drops sharply as the noise rate increases, which could be because misaligned sample pairs destroy the reliability of the supervised information. Thanks to robust learning from noisy pseudo labels, our ROLL achieves the best performance under several noise rates, which shows its effectiveness and stability. (3) According to the performance curves, the proposed robust noise pseudo-label learning framework consistently maintains performance advantages at different noise rates, even under the extreme 100% noise condition. This further demonstrates the effectiveness and robustness of our ROLL in handling multi-view data with noisy correspondence.

4.5. Visualization

To show the superiority of our method, on the CUB dataset with 50% noise rate, we adopt t-SNE to visualize the learned representations of RMCNC (the SOTA method designed specifically for NCP), MVCAN (the SOTA method designed specifically for practical noise-view scenario), and our ROLL, respectively. As shown in Fig.4, it is obvious that ROLL exhibits more compact cluster distributions and fewer outliers. Compared with MVCAN, we design a robust multi-view contrastive loss to alleviate the negative influence of false correspondences. Different from RMCNC, our ROLL can robustly learn from noisy pseudo-labels that can provide stronger semantic supervision.

4.6. Parameter sensitivity analysis

To evaluate the impacts of the two hyper-parameters α and β, we adjust their values to conduct parameter sensitivity analysis on the Scene15 dataset with 20% noise rate. Specifically, we vary their values within the range of [1e-5, 1e-4, 1e-3, 1e-2, 1e-1]. The corresponding performance curves of three metrics are shown in Fig.5. We can clearly observe that α andβ yield the optimal clustering results when they are around 1e-4 and 1e-2, respectively. If α is too small, our method cannot explore multi-view consistency. Once α is too large, the impact of correspondence noise could increase, thereby resulting in a significant drop



(a) ACC (b) NMI (c) ARI

Figure 5. Parameter analysis on Scene15 with 20% noise rate.

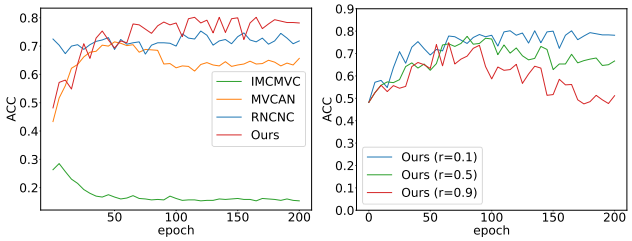


Figure 6. Robustness analysis on CUB with 20% noise rate.

in performance. Additionally, when β is greater than 1e-2, our ROLL cannot learn semantic supervised information from noisy pseudo-labels. As the value of β is close to 1e-2, the clustering performance is slightly affected, which indicates that the appropriate β value can improve the robustness of the model. In general, these results reflect the importance of each module from our ROLL.

4.7. Robustness Analysis

To show the influence of the r value on robustness, we perform a series of experiments on the CUB dataset with 20% noise rate. As shown in Fig.6, we draw the performance curves of several SOTA methods (ICMVC, MVCAN, RMCNC) and different r values ($r = 0.1, 0.5, 0.9$). Compared with several SOTA methods, our ROLL shows the best performance due to avoiding the overfitting and underfitting issues. When the r value is small, ROLL has better robustness. On the contrary, when the r value is larger, ROLL could overfit the false supervision, thereby resulting in a sharp drop in the clustering performance.

4.8. Ablation Analysis

To study the impact of each component of our ROLL on the clustering performance, we perform ablation experiments on the Scene15 and CUB datasets with 20% noise rate. Specifically, as shown in Tab.2, we compare the proposed ROLL with its eight variants, where CE and CL represent replacing $\mathcal{L}_{npl}$ and $\mathcal{L}_{rcl}$ with Eq.4 and Eq.9, respectively. According to the experimental results, we could draw the following conclusions: (1) When the RCL module is removed, the clustering performance of our model remarkably degrades, which indicates that RCL could enhance the robustness of neural networks against NCP. (2) When we remove the NPL module, due to the lack of clear semantic guidance, our variant has difficulty in effectively exploring multi-view semantic information through an unsupervised manner, thus weakening the discriminative ability. (3) The results of the last two variants show that our ROLL has better clustering performance and robustness.

Table 2. Ablation studies (%) of our ROLL on two datasets.

$\mathcal{L}_{rec}$	$\mathcal{L}_{npl}$	$\mathcal{L}_{rcl}$	Scene-15			CUB		
			ACC	NMI	ARI	ACC	NMI	ARI
	✓		18.29	23.97	7.95	49.87	54.46	35.46
		✓	43.93	40.94	25.03	60.53	54.67	41.96
	✓	✓	43.52	40.85	25.27	67.03	58.16	46.30
✓			33.42	34.91	17.99	40.00	30.29	18.42
✓	✓		35.20	37.19	19.49	43.93	37.36	22.59
✓		✓	44.76	45.42	27.78	79.57	76.24	66.25
✓	CE	✓	43.31	43.90	26.73	77.93	75.31	65.21
✓	✓	CL	35.13	34.77	18.50	35.50	26.04	14.06
✓	✓	✓	**45.65**	**45.00**	**28.15**	**81.03**	**76.81**	**68.11**

5. Conclusion

In this paper, to address the negative effects of NPP and NCP, we propose a novel pseudo-label guided multi-view learning framework, namely ROLL. Our ROLL first utilizes contrastive learning to warm up the model in a self-supervised manner, thereby obtaining the pseudo-labels of each view. To learn discrimination representations from the imperfect predictions, we construct the cross-view correspondence weights to establish cross-view relationships, thereby achieving noise-tolerance pseudo-label learning. Moreover, we propose robust multi-view contrastive learning to mitigate the overfitting problem caused by NCP. Extensive experiments demonstrate that ROLL outperforms 11 state-of-the-art MVC methods under different ratios.

Acknowledgments

This work was supported in part by the National Key R&D Program of China (Grant No. 2024YFB4710604), the NSFC (Grant No. 62472295, and 62372315), the Sichuan Science and Technology Planning Projects (Grant No. 2024NSFTD0047, 2024YFHZ0089, 2024NSFTD0049, and 2024YFHZ0144), the System of Systems and Artificial Intelligence Laboratory pioneer fund grant, the Fundamental Research Funds for the Central Universities (Grant No. CJ202403), the Chengdu Science and Technology Project (Grant No. 2023-XT00-00004-GX).

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